Category

Mortgage-Backed Security (MBS) Instrument and Market Features

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TIME TRANCHING

Learning Outcome

- define prepayment risk and describe time tranching structures in securitizations and their purpose

ABS and MBS pass-through security cash flows are uncertain because scheduled and actual payments frequently differ. Borrower behavior, such as the early payment of interest and principal as well as payment delinquencies over the life of a loan, depends upon a number of factors, including the income of the borrower (household income or cash flows for a corporation), the terms and conditions of the loan (such as the original interest rate versus current rates and any prepayment penalties), and the sale of the underlying asset being financed. To correctly value these securities, investors make certain assumptions about the predicted contractual payments after factoring in prepayment risk. **Prepayment risk** is the risk that the principal or a proportion of the principal is paid back at a different pace from the contractually agreed scheduled payment plan by the borrower. Prepayment risk has two components: contraction risk and extension risk, both of which reflect changes in the general level of interest rates.

Prepayment Risk

When interest rates decline, actual prepayments for fixed-rate mortgages will be higher than forecasted because homeowners will refinance at lower interest rates. Mortgage refinancing is particularly common in the US given the absence of loan prepayment penalties, which are more prevalent in other countries. Thus, the maturity of an MBS will be shorter than what was anticipated at the time of purchase. **Contraction risk** is the risk that the borrower repays the principal or a portion of the principal in a shorter period of time than the contractually agreed scheduled payment, reducing the amount of future payments the investor receives. For investors, this has two adverse consequences. First, investors must reinvest the proceeds at lower interest rates. Second, the prepayment option reduces the potential price appreciation for the bond.

When interest rates increase, actual prepayments will be lower than forecasted because homeowners will become less likely to refinance their mortgages and may even delay new home purchases. Thus, the maturity of an MBS will be longer than what was anticipated at the time of purchase. **Extension risk** is the risk that the borrower repays the principal or a proportion of the principal in a longer time period than the contractually agreed scheduled payment. For investors, this has adverse consequences: Higher interest rates reduce the value of the cash flows investors receive. Therefore, payments the investors receive will be discounted at a higher interest rate, and the extension stretches out the payments the investors receive.

The structure of a securitization may also allow the redistribution of “prepayment risk” among bond classes. An approach for reducing this risk is to create bond classes that possess different expected maturities. This is referred to as **time tranching**. For instance, a securitization pool may contain sequential tranching, where the principal repayments flow first to one tranche until the principal is fully repaid for that tranche and then to the sequential tranche until the principal is repaid for that tranche.

EXAMPLE 1

Time Tranching with Waterfall for RLT MBS

investors could face prepayment or extension risk in this instance. The specific maturity dates are noted in Exhibit 1.

Exhibit 1:

Overview of Notes Issued by Residential Loan Trust – Brief Summary of Maturity Terms

Maturity of Mortgage-Backed Notes Issued by the Residential Loan Trust	
Issuer:	Residential Loan Trust (RLT)
Status and Ranking of Payments	**Tranches**
Class A Notes rank senior to the Class B, C, & D Notes.	Tranche A Notes
	Face value: EUR825 million Interest rate: MRR + 0.50% Maturity: One year
Class B Notes rank senior to Class C & D Notes and rank junior to Class A Notes.	Tranche B Notes
	Face value: EUR100 million Interest rate: MRR + 1.50% Maturity: Two years
Class C Notes rank senior to the Class D Notes and rank junior to Class A & B Notes.	Tranche C Notes
	Face value: EUR50 million Interest rate: MRR + 2.50% Maturity: Three years
Class D Notes will rank junior to Class A, B, & C Notes.	Tranche D Notes
	Face value: EUR25 million Interest rate: Variable Maturity: Four years
Principal Payment:	To be paid on [Maturity Date]
Seniority:	The Notes are secured obligations of RLT.
Business Days:	Frankfurt

For the investor purchasing EUR5 million across Tranche B and Tranche C notes, loan prepayments could impact principal repayment as noted below:

Timing of Principal Repaid			
Investor's Initial RLT MBS Purchases		When principal is repaid if no prepayments in the RLT collateral pool	When principal is repaid if interest rates decline early in year two, thereby driving immediate prepayments in the RLT pool
Class A Notes	EUR0 million	One year, per MBS terms	One year, per MBS terms
Class B Notes	EUR3 million	Two years, per MBS terms	Less than two years
Class C Notes	EUR2 million	Three years, per MBS terms	Less than three years
Class D Notes	EUR0 million	Four years, per MBS terms	Less than four years

We would require additional information about the value of all prepayments and their exact timing to identify the exact principal repayment date.

As seen in Exhibit 1, it is possible and quite common for a securitization to have structures with both credit tranching (subordinated structures) and time tranching (different expected maturities).

Volkswagen Emissions Scandal

Investors are exposed to certain types of idiosyncratic risks when buying asset-backed securities (ABS). For buyers of auto ABS, major automaker recalls are a known risk to bond investors. In 2015, the US Securities and Exchange Commission (SEC) accused Volkswagen (VW), one of the world's largest carmakers, and its CEO Martin Winterkorn with defrauding American investors during an emissions scandal (the "Dieselgate" scandal). Volkswagen later admitted that it cheated on diesel emission tests, and this admission wiped billions of euros from the company's market value and led to the resignation of Winterkorn. The emissions scandal that rocked Volkswagen also impacted ABS tied to car loans and leases. Sales of auto ABS were booming in the years just prior to the scandal as investors sought higher-yielding products given the low interest-rate environment.

About USD5.6 billion worth of VW auto ABS were outstanding in 2015, with USD4.39 billion coming from bonds backed by loans and leases. Volkswagen, facing potential fines and litigation, found its ability to attract new business temporarily crimped, forcing down the values of the cars backing such loans. There were USD1.25 billion worth of bonds that were backed by car dealer inventories of VW cars, known as dealer floorplan ABS. For dealer floorplan ABS, the ABS trust benefits from VW financing assistance, including but not limited to VW's pledge to repurchase unsold new vehicles and inventory. Car dealers, told to immediately halt sales of popular 2015 and 2016 models, including Volkswagen's Jetta and Beetle convertible, were saddled with unsalable product. Dealers had a number of cars that they could not sell, so inventory did not turn over as rapidly. The ABS most adversely affected by VW's sales stoppage of certain 2015 and 2016 diesel models was the dealer floorplan securitization.

Debt investors were demanding greater compensation from Volkswagen after the company said its emissions-cheating scandal would cost the firm more than EUR6.5 billion. Volkswagen, a frequent issuer in Europe's asset-backed securities market, said that EUR2.6 billion of loans or leases for vehicles built with engines using cheating software were packaged into its securitizations.

Late in 2015, Volkswagen was marketing a new auto ABS with a yield premium that was three times bigger than a similar deal earlier that year. The German carmaker was offering to pay about 60 bps to 65 bps more than benchmark rates for top-ranking notes of the securitization of auto leases. In comparison, Volkswagen paid a spread of 20 bps over the one-month euro interbank offered rate for similar debt in April 2015.

Question Set

1. Creating bond classes that possess different expected maturities is referred to as:

A. subordination.
B. time tranching.
C. credit tranching.

Hide Solution

Solution:

The correct answer is B. An approach for reducing "prepayment risk" or "extension risk" among bond classes is to create bond classes that possess different expected maturities. This is referred to as time tranching. A and C are incorrect because subordination, also commonly referred to as a "waterfall"

2. Identify which of the following statements regarding securitized products is most accurate:

A. With time tranching, the tranches of a CMO amortize over time and all tranches mature at the same time.
B. Pass-through securities are the simplest securitized structure and involve issuers retaining the underlying assets.
C. In a true securitization, the underlying pool of assets is removed from the balance sheet and transferred into an independent legal entity that issues securities backed by these assets.

Hide Solution

Solution:

The correct answer is C. In a true securitization, the specific pool of assets is removed from the balance sheet and transferred into a separate and independent legal entity that then issues securities backed by these pooled assets. A is incorrect because time tranching is an approach to reduce prepayment or extension risk by creating bond classes that possess different expected maturities. B is incorrect because covered bonds, not pass-through securities, are the simplest securitization structure; they involve the issuer retaining the underlying assets on its balance sheet.

3. Issuing the collateralized bond classes with different expected maturities is *most likely* designed to mitigate:

A. credit risk.
B. default risk.
C. prepayment risk.

Hide Solution

Solution:

The correct answer is C. Prepayment risk is the uncertainty that the cash flows will be different from the scheduled cash flows as set forth in the lease agreement because of the lessors' ability to alter payments. The creation of bond classes with different expected maturities, referred to as time tranching, allows prepayment risk to be redistributed among bond classes. A is incorrect because credit risk pertains to the risk of default in the assets backing the collateralized bonds, not to the timing of payments, and B is incorrect because default risk again pertains to the risk of default in the assets backing the collateralized bonds.

4. *Determine the correct answers to complete the following sentences:*

_____ is the risk that the borrower might pay back the money borrowed more quickly than anticipated, reducing the amount of future payments the investor receives. _____ is the risk that the borrower might pay back the money borrowed more slowly than anticipated, extending the time of repayment and the maturity of the bond.

Hide Solution

Solution:

Contraction risk is the risk that the borrower might pay back the money borrowed more quickly than anticipated, reducing the amount of future payments the investor receives. *Extension risk* is the risk that the borrower might pay back the money borrowed more slowly than anticipated, extending the time of repayment and the maturity of the bond.

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